

LAURA LÓPEZ GALDO

Computational Neuroscientist · Biomedical Engineer · Neural Data Scientist

Biomedical engineer with a PhD in computational neuroscience, specializing in **machine learning** and **signal processing** for **high-dimensional neural data**. Experienced in building scalable analysis pipelines, extracting structure from complex time-series data, and developing predictive and interpretable models.

✉ lauralg_burela@hotmail.es  github.com/lauralopezgaldo  /in/laura-lopez-galdo

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, scikit-learn, Pytorch) MATLAB	Data Science & ML	Dimensionality reduction (PCA, LDA, t-SNE, NMF) Supervised & unsupervised learning Statistical modeling & inference
Signal Processing	Time-frequency analysis Spectral decomposition & burst detection algorithms Multivariate signal analysis	Animal experimentation	Project Designer level - non-human primate models

EXPERIENCE

01/2026 - Present	Postdoctoral Researcher · Neural Data Analysis	Université de Strasbourg (FR)
	- Developed signal processing pipelines to analyze oscillatory dynamics in neural data. - Applied decomposition methods to identify spatio-temporal patterns in multi-channel recordings. - Investigated coordination of signals across layers to extract task-relevant features.	
10/2021 - 12/2025	Early Stage Researcher within In2PrimateBrains (Marie Curie Fellow)	Aix-Marseille Université (FR)
	- Built machine learning pipelines to analyze high-dimensional neural datasets (<i>i.e.</i>, ~30 simultaneous laminar channels x ~50 cortical sites x multiple behavioral conditions). - Applied dimensionality reduction techniques to uncover low-dimensional structure in complex data. - Developed models to track temporal evolution of latent representations during task execution. - Quantified relationships between neural activity and behavioral variables.	
11/2020 - 04/2021	Research intern at the Neuromuscular Diagnostics Lab	Technical University of Munich (DE)
	<i>Modeling context-dependent encoding of motor behavior</i> - Designed and analyzed experiments generating behavioral datasets under controlled perturbations. - Developed computational models to simulate and compare alternative hypotheses. - Evaluated model performance by comparing predictions with empirical data.	
10/2017 - 07/2018	Research intern at the Robotics Lab	Universidad Carlos III de Madrid (ES)
	<i>Development of a Soft Hand Exoskeleton with Shape Memory Alloy Actuation</i> - Designed and prototyped a wearable SMA-based hand exoskeleton for rehabilitation. - Built custom 3D-printed molds and silicone structures for patient-specific fitting. - Developed a MATLAB-Simulink control system with force feedback for precise actuation and grip.	

EDUCATION

10/2021 - 12/2025	PhD in Computational Neuroscience	Aix-Marseille Université (FR)
	Early stage researcher within In2PrimateBrains, a Marie Skłodowska-Curie Innovative Training Network.	
10/2018 - 06/2021	MSc. Neuroengineering	Technical University of Munich (DE)
09/2014 - 07/2018	BSc. Biomedical Engineering	Universidad Carlos III de Madrid (ES)

PUBLICATIONS

2026	Coordinated multilaminar dynamics underlie multiplexed computation in macaque motor cortex	<i>under review at Nat. Commun.</i>
	López-Galdo, L., Nougaret, S., Battaglia D.*, & Kilavik, B. E.* DOI: 10.1101/2025.11.26.690660	
2025	Laminar dynamics in the macaque motor cortex	Theses.fr
	López-Galdo, L. https://theses.fr/s381148	
2024	Low and high beta rhythms have different motor cortical sources and distinct roles in movement control and spatiotemporal attention	PLOS Biology
	Nougaret, S., López-Galdo, L., ..., & Kilavik, B. E. DOI: 10.1371/journal.pbio.3002670	
2026	Area- and Layer-Specific Organization of Multimodal Timescales in Macaque Motor Cortex	bioRxiv
	Nandi, N., López-Galdo, L., Nougaret, S., & Kilavik, B. E. DOI: 10.64898/2026.03.21.713374	

SCIENTIFIC COMMUNICATION

Experience in communicating complex concepts to technical and non-technical audiences at international events.

Posters

- Cosyne 2026 - Portugal
- NeuroFrance 2025 - Montpellier
- FENS Forum 2024 - Vienna
- GDR NeuralNet 2024 - Paris
- Society for Neuroscience 2023 - Washington D.C.

Talks

- GDR NeuralNet 2023 - Marseille | *Selected talk*
- Thematic Workshops by In2PrimateBrains Consortium | Rome 2022, Heraklion 2023, Newcastle 2024

Public Outreach

- Talk at IBS of Provence (2023 & 2024)
A journey to the science of the brain
- Scientific event in Newcastle University Campus (2024)

PEER REVIEW

- Review activity for **Nature Communications** in 2025 under the *Early Career Researcher Co-Review Initiative*.

SCHOLARSHIPS AND AWARDS

2024	Neuroschool End-of-PhD funding grant Six-month scholarship to fund the end of the research project carried out during the PhD.	Neuromarseille Institute
2021 - 2024	Marie Skłodowska-Curie Actions - Horizon 2020 I got European Union funding for 36 months as an early stage researcher to develop my PhD project.	Aix-Marseille Université
2017	Academic Excellence Scholarship Award for academic excellence given by the Community of Madrid to undergraduate students.	Comunidad de Madrid
2016 - 2017	Scholarship for Non-European Mobility Program Scholarship covering university fees for studying one academic year of biomedical engineering in the US.	Universidad Carlos III de Madrid

LANGUAGES

Spanish - native, **Galician** - native, **English** - C1, **French** - B2/C1, **Greek** - beginner